

types of personas based on a given template. Each persona needs to have information including name, age, gender, address, job and background. Examples of the personas are shown in Appendix H.1.

Belief-Desire-Intention (BDI). The BDI framework is a well-established approach in agent-oriented programming (Rao et al., 1995) and was recently adopted to language models (Andreas, 2022). We propose modeling LLM agents in Trust Games with the BDI framework to gain deeper insights into LLM agents’ behaviors. Specifically, we let LLM agents directly output their Beliefs, Desires, and Intentions as the reasoning process for decision-making in Trust Games.

3 Do LLM Agents Manifest Trust Behavior?

In this section, we investigate whether or not LLM agents manifest trust behavior by letting LLM agents play the Trust Game (Section 2.1 Game 1). In Behavioral Economics, trust is widely measured by the initial amount sent from the trustor to the trustee in the Trust Game (Glaeser et al., 2000; Cesarini et al., 2008). Following the measurement of trust in human studies and the assumption humans own reasoning processes that underlie their decisions, we can define the conditions that LLM agents manifest trust behavior in the Trust Game as follows. *First, the amount sent is positive and does not exceed the amount of money the trustor initially possesses*, which implies that the trustor places self-interest at risk with the expectation the trustee will reciprocate and that the trustor understands the money limit that can be given. *Second, the decision (i.e., amounts sent) can be interpreted as the reasoning process (i.e., the BDI) of the trustor.* We explored utilizing BDI to model the reasoning process of LLM agents. If we can interpret the decision as the articulated reasoning process, we have evidence that LLM agents do not send a random amount of money and manifest some degree of rationality in the decision-making process. Then, we assess whether LLM agents exhibit trust behavior based on two aspects: the amount sent and the BDI.

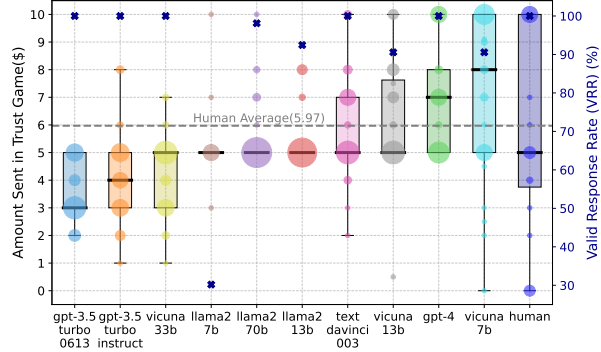


Figure 2: **Amount Sent Distribution of LLM Agents and Humans as the Trustor in the Trust Game.** The size of circles represents the number of personas for each amount sent. The bold lines show the medians. The crosses indicate the VRR (%) for different LLMs.

3.1 Amount Sent

To evaluate LLMs’ capacity to understand the basic experimental setting regarding money limits, we propose a new evaluation metric, Valid Response Rate (VRR) (%), defined as the percentage of personas with the amount sent falling within the initial money (\$10). Results are shown in Figure 2. We can observe that **most LLMs have a high VRR except Llama-7b**, which implies that most LLMs manifest a full understanding regarding limits on the amount they can send in the Trust Game. Then, we observe the distribution of amounts sent for different LLMs as the trustor agent and discover that **the amounts sent are predominantly positive, indicating a level of trust.**

3.2 Belief-Desire-Intention (BDI)

The sole evidence of the amount sent cannot sufficiently support the existence of trust behavior, because agents could send positive but random amounts of money. Thus, we leveraged the Belief-Desire-Intention framework (Rao et al., 1995; Andreas, 2022) to model the reasoning process of LLM agents. If we can interpret the amounts sent from BDI outputs, we have evidence to refute the hypothesis that the amounts sent are positive but random and demonstrate that LLM agents manifest some degree of rationality. We take GPT-4 as an example to analyze its BDI outputs. More examples from the other nine LLMs such as Vicuna-v1.3-7b are shown in the Appendix I. Considering that the amounts sent typically vary across distinct personas, we select one BDI from the personas that give a high amount of money and another BDI from those that give a low amount. **Positive** and **negative** factors for trust behavior in the reasoning process are marked in blue and red, respectively.

As a person with a strong belief in the goodness of humanity, I trust that the other player ...Therefore, my desire is to maximize the outcome for both of us and cement a sense of com-

radery and trust... I intend to use this as an opportunity to add what I can to someone else's life...Finally, I will give **10 dollars**.

We can observe that this persona shows a high-level of “comradery and trust” towards the other player, which justifies the high amount sent from this persona (*i.e.*, **10 dollars**).

As an Analyst,... My desire is that the other player will also see the benefits of reciprocity and goodwill ... my intention is to give away a significant portion of my initial 10 ... However, since I have no knowledge of the other player, ... Therefore, I aim to give an amount that is not too high, ...Finally, I will give **5 dollars** to the other player..

Compared to the first persona, we see that the second one has a more cautious attitude. For example, “since I have no knowledge of the other player” shows skepticism regarding the other player’s motives. Thus, this persona, though still optimistic about the other player (“intention ... give away a significant portion”), strategically balances risk and reciprocity, and then decides to send only a modest amount.

Based on GPT-4’s BDI examples and examples from other LLMs in Appendix I, we find **decisions (*i.e.*, amounts sent) from LLM agents in the Trust Game can be interpreted from their articulated reasoning process (*i.e.*, BDI)**. Because most LLM agents have a high VRR—send a positive amount of money—and show some degree of rationality in giving money, our first core finding is:

Finding 1: LLM agents generally exhibit trust behavior under the framework of the Trust Game.

3.3 Basic Analysis of Agent Trust

We also conduct a basic analysis of LLM agents’ trust behavior, namely agent trust, based on the results in Figure 2. *First*, we observe that Vicuna-7b has the highest level of trust towards the other player and GPT-3.5-turbo-0613 has the lowest level of trust as trust can be measured by the amount sent in human studies (Glaeser et al., 2000; Cesarini et al., 2008). *Second*, compared with humans’ average amount sent (\$5.97), most personas for GPT-4 and Vicuna-7b send a higher amount of money to the other player, and most personas for LLMs such as GPT-3.5-turb-0613 send a lower amount. *Third*, we see that amounts sent for Llama2-70b and Llama2-13b have a convergent distribution while amounts sent for humans and Vicuna-7b are more divergent.

4 Does Agent Trust Align with Human Trust?

In this section, we aim to explore the fundamental relationship between agent and human trust, *i.e.*, whether or not agent trust aligns with human trust. This provides important insight regarding the feasibility of utilizing LLM agents to simulate human trust behavior as well as more complex human interactions that involve trust. First, we propose a new concept *behavioral alignment* and discuss its distinction from existing alignment definitions. Then, we conduct extensive studies to investigate whether or not LLM agents exhibit alignment with humans regarding trust behavior.

4.1 Behavioral Alignment

Existing alignment definitions predominantly emphasize *values* that seek to ensure the safety and helpfulness of LLMs (Ji et al., 2023; Shen et al., 2023; Wang et al., 2023c), which cannot fully characterize the landscape of multifaceted alignment between LLMs and humans. Thus, we propose a new concept of *behavioral alignment* to characterize the LLM-human analogy regarding *behavior*, which involves both actions and the associated reasoning processes that underlie them. Because actions evolve over time and the reasoning that underlies them involves multiple factors, we define *behavioral alignment* as the analogy between LLMs and humans concerning factors impacting behavior, namely *behavioral factors*, and action dynamics, namely *behavioral dynamics*.

Based on the definition of behavioral alignment, we aim to answer: *does agent trust align with human trust?* As for *behavioral factors*, existing human studies have shown that three basic factors impact human trust behavior including reciprocity anticipation (Berg et al., 1995; Cox, 2004), risk perception (Bohnet & Zeckhauser, 2004) and prosocial preference (Alós-Ferrer & Farolfi, 2019). We examine whether agent trust aligns with human trust along these three factors. Although *behavioral dynamics* vary for different humans and agent personas, we analyze whether agent trust has the same patterns across multiple turns as human trust in the Repeated Trust Game.

Besides analyzing the trust behavior of LLM agents and humans based on quantitative measurements (*e.g.*, the *amount sent* from trustor to trustee), we also explore the use of *BDI* to interpret the reasoning

process with which LLM agents justify their actions, which can further validate whether LLM agents manifest an underlying reasoning process analogous to human cognition.

4.2 Behavioral Factor 1: Reciprocity Anticipation

Reciprocity anticipation, the expectation of a reciprocal action from the other player, can positively influence human trust behavior (Berg et al., 1995). The effect of reciprocity anticipation exists in the Trust Game but not in the Dictator Game (Section 2.1 Games 1 and 2) because trustee cannot return money in the Dictator Game, which is the only difference between these games. Thus, to determine whether LLM agents can anticipate reciprocity, we compare their behaviors in these Games.

First, we analyze trust behaviors based on the average amount of money sent by human or LLM agents. As shown in Figure 3, human studies show that humans exhibit a higher level of trust in the Trust Game than in the Dictator Game (\$6.0 vs. \$3.6, p -value = 0.01 using One-Tailed Independent Samples t-test) (Cox, 2004), indicating that reciprocity anticipation enhances human trust. Similarly, GPT-4 (\$6.9 vs. \$6.3, p -value = 0.05 using One-Tailed Independent Samples t-test) also shows a higher level of trust in the Trust Game with statistical significance, implying that reciprocity anticipation can enhance agent trust. However, LLMs with fewer parameters (e.g., Llama2-13b) do not show this tendency in their trust behaviors for the Trust and Dictator Games.

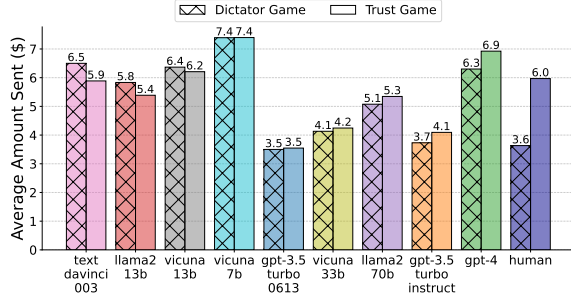


Figure 3: The Comparison of Average Amount Sent for LLM Agents and Humans in the Trust Game and the Dictator Game.

Then, we further analyze GPT-4 agents’ BDI to explore whether they can anticipate reciprocity in their reasoning (the complete BDIs are in Appendix 1.10). Typically, in the Trust Game, one persona’s BDI emphasizes “*putting faith in people*”, which implies the anticipation of the goodness of the other player, and “*reflection of trust*”. However, in the Dictator Game, one persona’s BDI focuses on concepts such as “*fairness*” and “*human kindness*”, which are not directly tied to trust or reciprocity. Thus, we can observe that GPT-4 shows distinct BDI outputs in the Trust and Dictator Games.

Based on the above analysis of the amount sent and BDI, we find that **GPT-4 agents exhibit human-like reciprocity anticipation in trust behavior**. Nevertheless, **LLMs with fewer parameters (e.g., Llama2-13b) do not show an awareness of reciprocity from the other player**.

4.3 Behavioral Factor 2: Risk Perception

Existing human studies have demonstrated the strong correlation between trust behavior and risk perception, suggesting that human trust will increase as risk decreases (Hardin, 2002; Williamson, 1993; Coleman, 1994). We aim to explore whether LLM agents can perceive the risk associated with their trust behaviors through the MAP Trust Game and the Risky Dictator Game (Section 2.1 Games 3 and 4), where risk is represented by the probability $(1-p)$ (defined in Section 2.1).

As shown in Figure 4, we measure human trust (or agent trust) by the portion choosing to trust the other player in the whole group, namely the Trust Rate (%). Based on existing human studies (Bohnet & Zeckhauser, 2004), when the probability p is higher, the risk for trust behaviors is lower, and more humans choose to trust, manifesting a higher Trust Rate, which indicates that human trust rises as risk falls. Similarly, we observe a general increase in agent trust as risk decreases for LLMs including GPT-4, GPT-3.5-turbo-0613, and text-davinci-003. In particular, we can see that the curves of humans and GPT-4 are more

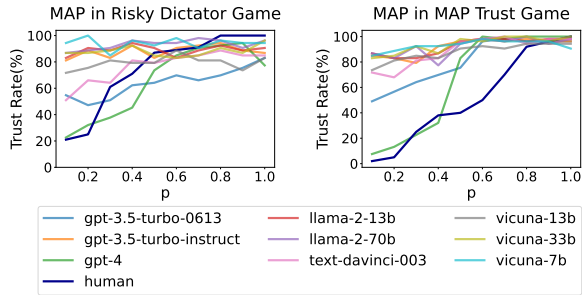


Figure 4: Trust Rate (%) Curves for LLM Agents and Humans in the MAP Trust Game and the Risky Dictator Game. The metric Trust Rate indicates the portion of trustors opting for trust given p .